

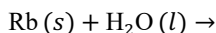
Name: _____

These questions assess your understanding of the material from Chapters 3, 4 of Openstax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Julia weighs out 0.9966 g of hypochlorous acid (HClO) into a 500.-mL volumetric flask and dilutes the solution to the mark with distilled water. She plans to titrate the acid with 0.3729-M zinc hydroxide solution. Calculate the volume of zinc hydroxide solution that Julia must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

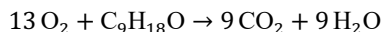
ANSWER: _____

2. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".



ANSWER: _____

3. What is the total number of carbon atoms shown on the product side of this chemical equation?



ANSWER: _____

4. A strip of solid cobalt metal is put into a beaker of 0.048-M NaBrO solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

5. Each week, Deskins Chemicals, Inc., a manufacturer of fine chemicals, uses 173 kg of compound A and 121 kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 73%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

6. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 63.8 g.

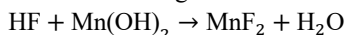
Compound	Initial Mass	Final Mass
A	64.0 g	0 g
B	52.4 g	35.6 g

ANSWER: _____

7. Nicholas conducted some measurements and determined that 0.238 moles of bromine liquid participated in a chemical reaction. Calculate the mass of bromine liquid that participated in the chemical reaction.

ANSWER: _____

8. Given this unbalanced chemical equation, calculate and report the percent yield of MnF_2 if 4.59 g of MnF_2 is produced from the reaction of 7.54 g of HF and 9.11 g of Mn(OH)_2 . Ensure that your answer has the correct number of significant figures.

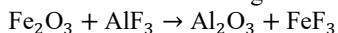


ANSWER: _____

9. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

10. Given this unbalanced chemical equation, report the mass of Fe_2O_3 needed to produce 3.534 g of FeF_3 . Ensure your answer has the correct number of significant figures.



ANSWER: _____

11. The chemical formula for calcium oxide is CaO . Calculate the molar mass of calcium oxide. Round your answer to 4 significant figures.

ANSWER: _____

12. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

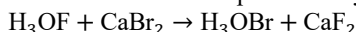
13. As part of his routine quality-control checks, Kyle, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of chlorous acid solution. He takes a 50.0-mL sample of chlorous acid solution and titrates it with a copper (II) hydroxide solution, which he made by dissolving 2.55 g of solid copper (II) hydroxide in 1.37 L of distilled water. If Kyle finds that it takes 272.3 mL of the copper (II) hydroxide solution for the titration to reach the equivalence point, what molarity of the chlorous acid solution does Kyle calculate?

ANSWER: _____

14. Christopher is instructed to dissolve 2.242 g of chromium (III) chloride in 143 mL of a 26.0-mM aqueous solution of hydronium sulfide. Calculate the molarity of the chromium (III) cation in the solution after the reaction has occurred. Assume the volume of the solution does not change when chromium (III) is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

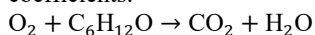
15. A way to produce solid calcium fluoride is by combining aqueous hydronium fluoride with aqueous calcium bromide. The chemical reaction is represented by the following unbalanced chemical equation:



Knowing this, Shiv pours 13.79 mL of a 0.03081-M solution of aqueous hydronium fluoride and then adds 16.0 mL of a 0.13-M solution of aqueous calcium bromide. He then stirs the mixture and notices calcium fluoride precipitating out of the solution. What is the mass of calcium fluoride that he should expect the reaction to yield?

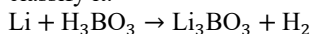
ANSWER: _____

16. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.



ANSWER: _____

17. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

18. An atom of copper has a mass of 1.654×10^{-22} g and a virus has a mass of 4.11×10^{-11} mg. How many moles of viruses have a mass equal to the mass of an atom? Ensure your answer has the correct number of significant figures.

ANSWER: _____

19. Cobalt (II) hydroxide reacts with boric acid to form cobalt (II) borate and water. Calculate the mass of boric acid needed to produce 12.19 g of cobalt (II) borate. Ensure your answer has the correct number of significant figures.

ANSWER: _____

20. Ammonium carbonate reacts with iron (III) chloride to form ammonium chloride and iron (III) carbonate. Suppose 9.19 g of ammonium carbonate is mixed with 7.63 g of iron (III) chloride. Calculate the maximum mass of ammonium chloride that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

21. Ashley has a 0.1055-M solution of sodium sulfide (Na_2S). If she pours 7.477 mL of sodium sulfide into a reaction flask, what is the mass of sodium sulfide that she has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: _____

22. Hydrochloric acid reacts with rubidium hydroxide to form rubidium chloride and water. Suppose 2.549 g of hydrochloric acid is mixed with 13.19 g of rubidium hydroxide. Calculate and report the theoretical yield of rubidium chloride formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

23. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

24. Sophia weighs out 0.928 g of an unknown triprotic acid into a 250.-mL volumetric flask and dilutes the solution to the mark with distilled water. She then titrates this solution with 0.360-M nickel (II) hydroxide solution. When the titration reaches the equivalence point, Sophia finds that she has added 30.702 mL of the nickel (II) hydroxide solution. Calculate the molar mass of the unknown triprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

25. Christopher must prepare 810. mL of a 0.859-mM working solution of aqueous hydronium phosphate ($(\text{H}_3\text{O})_3\text{PO}_4$). He will do this by pouring some 0.0203-M aqueous hydronium phosphate stock solution into a graduated cylinder and diluting it with distilled water. How much of the aqueous-hydronium phosphate stock solution must Christopher pour into the graduated cylinder? Report your answer in mL. Be sure your answer has the correct number of significant figures.

ANSWER: _____

26. As HNO_3 is converted into $\text{Ba}(\text{NO}_3)_2$ in the following reaction, determine whether oxygen is oxidized, reduced, or neither. Clearly explain why you chose your answer. Consider only the oxygen found in HNO_3 and $\text{Ba}(\text{NO}_3)_2$.
- $$\text{BaO} + \text{HNO}_3 \rightarrow \text{Ba}(\text{NO}_3)_2 + \text{H}_2\text{O}$$

ANSWER: _____

27. Sam prepares a solution of hydronium chloride (H_3OCl) by measuring out 273.5 mmol of hydronium chloride into a 25.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

28. Given this unbalanced chemical equation, report the amount of $(\text{NH}_4)_2\text{C}_2\text{O}_4$ needed to produce 0.6121 moles of $\text{Cs}_2\text{C}_2\text{O}_4$. Report your answer in moles. Ensure your answer has the correct number of significant figures.
- $$(\text{NH}_4)_2\text{C}_2\text{O}_4 + \text{Cs}_2\text{S} \rightarrow \text{Cs}_2\text{C}_2\text{O}_4 + (\text{NH}_4)_2\text{S}$$

ANSWER: _____

29. An atom of oxygen has a mass of 9.9×10^{-22} g and a particular planet has a mass of 7.52×10^{26} g. How many moles of oxygen atoms have a mass equal to the mass of this particular planet? Ensure your answer has the correct number of significant figures.

ANSWER: _____

30. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

31. Determine which of the following labels apply to the chemical compound H_2CO_3 : ionic, molecular, acid, and base.

ANSWER: _____

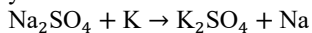
32. Nick must dilute 78.2 mL of a 6.035-mM solution of aqueous ammonium sulfate ($(\text{NH}_4)_2\text{SO}_4$) until the concentration falls to 1.374 mM. He will do this by adding distilled water to the solution. How much distilled water must he add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

33. A virus has a mass of 8.93×10^{-12} mg and a coin has a mass of 4.453×10^0 g. What is the mass of 1 mole of coins? Ensure your answer has the correct number of significant figures.

ANSWER: _____

34. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.



ANSWER: _____

35. Measurements show that an unknown compound has the following composition:

Element	Mass %
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Rubidium	91.4 %
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Oxygen	8.56 %
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Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

36. James conducted some measurements and determined that 522. g of nitrogen gas participated in a chemical reaction. Calculate the number of moles of nitrogen gas that participated in the chemical reaction.

ANSWER: _____

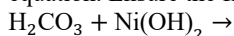
37. Calculate the number of carbon atoms in a 1.164×10^{-3} -g sample of tetraantimony tricarbide (Sb_4C_3).

ANSWER: _____

38. Calculate the mass percent of chromium in chromium (III) permanganate ($\text{Cr}(\text{MnO}_4)_3$). Round your answer to the nearest percentage.

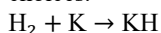
ANSWER: _____

39. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.



ANSWER: _____

40. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.



ANSWER: _____